

TOPIC 4: PLEIOTROPY, POLYGENIC INHERITANCE & DIHYBRID CROSS

PU-II Principles of Inheritance and Variation • Topic-Wise Worksheet

Student Name: _____

Roll / Reg. No: _____

Date / Batch: _____

SECTION A: Multiple Choice Questions (1 Mark Each)

- The phenotypic ratio obtained in the F₂ generation of a Mendelian dihybrid cross is:
(A) 9:3:3:1 (B) 1:2:1:2:4:2:1:2:1
(C) 1:1:1:1 (D) 3:1
- Which genetic phenomenon is exemplified by human skin colour and wheat kernel colour?
(A) Pleiotropy (B) Polygenic / Quantitative inheritance
(C) Multiple allelism (D) Complete linkage
- Starch grain size in pea seeds shows incomplete dominance in Bb heterozygotes, while seed shape shows:
(A) Co-dominance (B) Complete dominance (Round)
(C) Recessiveness (Wrinkled) (D) Polygenic inheritance
- Phenylketonuria (PKU) is an example of pleiotropy caused by mutation in the gene coding for:
(A) Tyrosinase (B) Phenylalanine hydroxylase
(C) Hexosaminidase A (D) Beta-globin

SECTION B: Fill in the Blanks (1 Mark Each)

- A single gene producing multiple phenotypic effects by influencing metabolic pathways is called a gene.
- Polygenic inheritance was experimentally demonstrated in wheat kernel colour by
- In human skin colour, the genotype with intermediate phenotype containing 3 dominant and 3 recessive alleles is
- Mendel's Law of Independent Assortment is based on the results of a cross.

SECTION C: Very Short Answer Questions (2 Marks Each)

9. [2M] State Mendel's Law of Independent Assortment.

10. [2M] What is Pleiotropy? Give two examples of pleiotropic genes in humans/plants.

SECTION D: Short Answer Questions (3 Marks Each)

11. [3M] Explain polygenic inheritance of human skin colour assuming three additive genes A, B, and C.

12. [3M] How does starch synthesis in pea seeds demonstrate both complete dominance and incomplete dominance?

SECTION E: Long Answer / Essay Questions (5 Marks Each)

13. [5M] Explain Mendel's Dihybrid Cross using seed shape (Round/Wrinkled) and seed colour (Yellow/Green) in *Pisum sativum*. Give Punnett square, phenotypic ratio, and genotypic ratio.

14. [5M] Write detailed notes on: (a) Chromosomal basis of independent assortment (b) Polygenic inheritance vs Multiple allelism with examples.